

DAY 02 — CYBER SECURITY

Operating Systems • Linux • Networking • Protocols • Security Fundamentals

Source: Clean study notes prepared from the Day 2 handwritten material provided.

1. Quick Fundamentals

Cookies: Small pieces of data stored by a website/browser to remember information about a user or session.

Session: A temporary interaction between a user and a server. It helps the server remember a user's activity while they are using an application.

Buffer: A temporary memory area used to hold data while it is being transferred between two places or devices.

Cache: A temporary storage area that keeps frequently used data so it can be accessed faster.

TCP: A connection-oriented transport protocol that provides reliable delivery of data.

UDP: A connectionless transport protocol that sends data without establishing a connection first.

2. Operating System Overview

An **Operating System (OS)** is system software that manages computer hardware and provides services for applications and users.

- Process Management — manages running programs/processes.
- Service Management — manages background services.
- Memory Management — manages and allocates memory.
- Searching Management — locating files/data/resources.
- Patching Management — applying updates and security fixes.
- Security Management — protects the system and controls access.
- Scripting Management — supports automation using scripts.
- Endpoint Security Management — protects end-user devices.
- Server Management — manages server resources and services.
- Double Sorting / Sorting — arranging data in an ordered form (as noted in the source material).

3. OS Components

The notes identify the following main OS-related components:

- Operating System
- Shell / Command Line Interface (CLI)
- Interpreter
- Compiler

- Hardware

4. Operating System Types and Booting

The notes mention Windows, Linux, macOS and Unix operating systems. The **booting process** is the sequence through which a computer starts and loads the operating system into memory.

Linux families mentioned:

- Debian family — Debian, Ubuntu, Kali Linux, etc.
- Red Hat family — Red Hat / RHEL, Fedora, CentOS and related distributions.

5. Linux Overview

Linux is an operating-system family commonly used on servers, desktops, cloud systems and security environments. The notes focus on the command-line interface (CLI) and basic Linux commands.

Basic Linux commands from the notes:

Command	Purpose
pwd	Shows the current working directory.
ls	Lists files and folders.
cd	Changes the current directory.
mkdir	Creates a directory.
touch	Creates an empty file.
cat	Displays file contents.
cp	Copies files or directories.
mv	Moves or renames files/directories.
rm	Removes files/directories.
clear	Clears the terminal screen.

6. Networking Fundamentals

The notes cover networking concepts such as network topology, IP addressing, protocols, ports, packet transfer and common network services.

Network Topologies:

- Star topology — devices connect to a central device such as a switch.
- Bus topology — devices share a common communication medium.
- Ring topology — devices are connected in a circular path.
- Tree topology — multiple star networks are connected in a hierarchy.
- Hybrid topology — combination of two or more topologies.

7. IP Addressing

IP address: A logical address used to identify a device on a network.

IPv4: Uses a 32-bit address, commonly written in dotted-decimal form, for example 192.168.1.10.

IPv6: Uses a 128-bit address and provides a much larger address space.

Private IP: Used inside private/local networks.

Public IP: Used for communication over the public Internet.

MAC address: A hardware/network-interface address used for communication at the data-link layer.

Subnetting: Dividing a network into smaller logical networks (subnets).

8. Common Protocols and Ports

Protocol	Port	Use
HTTP	80	Web traffic
HTTPS	443	Secure web traffic
FTP	21	File transfer
SSH	22	Secure remote login
Telnet	23	Remote login (unencrypted)
DNS	53	Domain-name resolution
DHCP	67/68	Automatic IP configuration
SMTP	25	Sending email
POP3	110	Receiving email
IMAP	143	Receiving/synchronizing email

9. TCP and UDP

TCP: Connection-oriented and reliable. It establishes a connection before transferring data and provides ordered delivery.

UDP: Connectionless and lightweight. It sends data without establishing a connection and is useful where speed is important.

Simple comparison: TCP = reliable delivery; UDP = faster, connectionless delivery.

10. Firewalls and Network Security

Firewall: A security mechanism that monitors and controls network traffic according to defined rules. It can allow or block traffic based on factors such as IP address, port and protocol.

Network security: Protects devices, networks and data from unauthorized access and attacks.

11. Storage / Memory / Performance Concepts

Cache: Stores frequently used data for faster access.

Buffer: Temporarily holds data during transfer or processing.

RAM: Main working memory used by running programs and processes.

Storage: Persistent space used to keep files and applications.

12. Security Concepts Noted in the Material

- Endpoint security — protecting computers and end-user devices.
- Application security — protecting applications from vulnerabilities and attacks.
- Network security — protecting network communication and infrastructure.
- Data security — protecting data from unauthorized access, modification or loss.
- Cloud security — protecting cloud systems, services and data.
- Email security — protecting email accounts and messages.

13. Quick Revision

- Cookie → remembers small pieces of website information.
- Session → temporary user-server interaction.
- Buffer → temporary holding area during data transfer.
- Cache → stores frequently used data for faster access.
- OS → manages hardware and software resources.
- Shell → interface for giving commands to the OS.
- IP → logical network address.
- MAC → network-interface hardware address.
- TCP → reliable, connection-oriented transport.
- UDP → connectionless transport.
- HTTP → port 80; HTTPS → port 443.
- SSH → port 22; DNS → port 53.
- Firewall → controls network traffic according to security rules.

End of Day 02 Notes

DAY 02 — CYBER SECURITY

Clear Notes from Your Handwritten Study Pages

Topics: OS connectivity, SSH, RDP, PuTTY, Linux shells, cloud, cookies, buffer, cache and OTT/video protocols.

1. Operating System Connectivity

The notes discuss connecting to different operating systems from another computer using remote-access tools and protocols.

Linux connectivity: A third-party tool such as **PuTTY** can be used to connect to a Linux system. The basic idea shown is: **Client** → **PuTTY** → **Linux**.

For command-line access, the notes use **SSH** with **TCP** and **port 22**.

SSH: Secure Shell, used for secure remote command-line access.

PuTTY: Third-party remote connection tool mentioned in the notes.

2. Windows Remote Connection

The notes mention **RDP (Remote Desktop Protocol)** for Windows remote access. RDP provides a graphical user interface (GUI) for the remote desktop.

RDP port: 3389 **Protocol type:** TCP.

The notes also mention Windows command-line mode and Windows tools/commands.

3. Mac OS Connectivity

Mac → **Mac:** SSH, TCP, port 22; RDP, port 3389, TCP (as written); SSH for command-line access.

Mac → **Linux:** SSH, port 22; xrdp, port 3389, TCP for graphical remote desktop access; SSH for command-line access.

4. Linux Shells

A **shell** is a command-line interface (CLI) that allows a user to communicate with the Linux operating system by typing commands.

sh — Bourne Shell — basic Unix/Linux scripting.

bash — Bourne Again Shell — common/default Linux shell and administration.

zsh — Z Shell — developers and interactive terminal use.

ksh — Korn Shell — enterprise/Unix scripting.

csh — C Shell — C-like syntax / older Unix environments.

tcsh — TENEX C Shell — enhanced C shell.

fish — Friendly Interactive Shell — beginner-friendly interactive shell.

5. Linux Commands / Tools Mentioned

ssh — secure remote command-line access.

bash — commonly used Linux shell.

PowerShell — Windows command-line/scripting environment mentioned in the notes.

win + R — Windows Run dialog shortcut mentioned in the notes.

RDP — Remote Desktop Protocol for graphical remote access.

xrdp — mentioned for remote desktop connectivity to Linux.

6. Cloud Computing Overview

The notes describe cloud as a collection of computing resources and services such as **database, storage, virtualization, configuration, security, identity and management (IAM)**, with scalable resources.

Cloud providers mentioned: AWS (Amazon Web Services), Microsoft Azure and Google Cloud (GCP).

A **region** is a geographic cloud location. Each region contains multiple **Availability Zones (AZs)**. The notes describe an AZ as an isolated data-center location.

Simple structure: Region → Availability Zones → Cloud resources/services.

7. What is a Cookie?

A **cookie** is a small piece of data that a website stores in your browser. It helps a website remember information about you or your activity.

Example: When you visit a website, some information may be stored in a cookie so the website can recognize or remember information later.

8. What is a Buffer?

A **buffer** is a temporary area used to hold data while it is being transferred from one place to another.

Example: While watching an online video, some video data may be temporarily stored in a buffer before it is played.

9. What is Cache?

A **cache** is temporary storage that keeps frequently used data so it can be accessed faster the next time.

The notes describe cache as storing frequently used information/data to improve speed and reduce repeated work.

10. Video Streaming / OTT Platforms

The notes ask which protocol, protocol type and port are used for services such as **YouTube, Amazon and Netflix / OTT platforms**.

The supplied notes record: **HTTP/HTTPS web communication, port 443, and TCP or UDP as the transport.**

Source-note form: Video streaming generally → HTTP/HTTPS → port 443 → transport TCP or UDP.

11. Quick Revision

- SSH → secure remote command-line access → port 22 → TCP (as noted).
- RDP → Windows remote desktop / GUI → port 3389 → TCP (as noted).
- PuTTY → third-party remote connection tool.
- Shell → CLI used to communicate with an operating system.
- Bash → Bourne Again Shell.
- Zsh → Z Shell.
- Cookie → small browser-stored data.
- Buffer → temporary holding area during data transfer.
- Cache → temporary storage for faster access.
- Cloud → collection of computing resources/services.
- Region → geographic cloud location.
- Availability Zone → isolated location/data-center area within a region.

Note: This document preserves the terminology and details from your handwritten pages. Where a page was unclear, I did not add unsupported details.

END OF DAY 02